

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Term Project Topic

From the syllabus: The term project is an independent ecological niche modeling analysis developed incrementally over the semester. Students will select a problem, compile and process occurrence and environmental data, build and evaluate one or more models, and interpret the results in an ecological context. The project will demonstrate a complete, reproducible workflow, including code, model outputs, and written interpretation of results, assumptions, and limitations. The term project will be evaluated based on the quality of the modeling workflow, including data selection and processing, model construction and evaluation, clarity of interpretation, and reproducibility of code and outputs. The final submission will include a written report, figures, and annotated R code.

The purpose of this first assignment for your term project is to identify a preliminary direction. Your topic can change or be refined as we work through data, scale, and modeling approaches this semester. Please submit a short description that includes:

- **Research question or idea** — What are you interested in asking?
- **Study organism/system** — What species or group are you considering?
- **Geographic scope** — Where would the project focus?
- **Biological motivation** — Briefly explain why your question is interesting or useful in ecological or conservation terms?
- **Any questions or uncertainties** — What would you like feedback on?

Format: PDF, up to 1 page, 12-point Times New Roman

Due: Friday, September 11 by 11:59 PM